



**Verified Carbon
Standard**

SOLITUDE 16MW SOLAR PV



Document Prepared by Voltas Yellow Ltd

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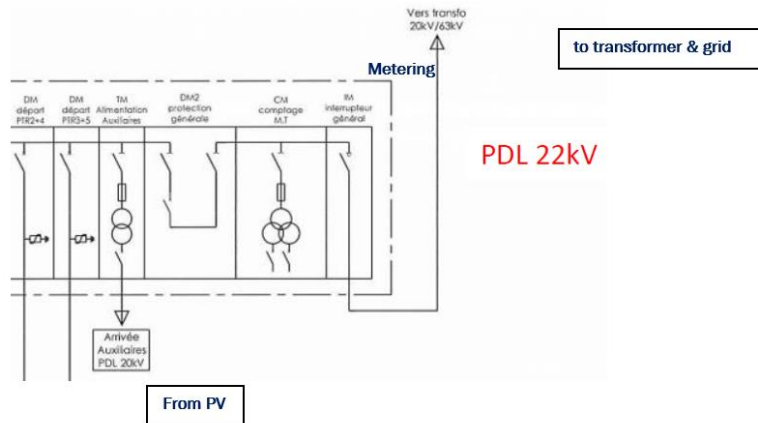
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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Solitude 16 MW solar PV (“the project”) consists of the construction and operation of a greenfield 16.328 MW_{DC} solar photovoltaic power plant in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses. It involves the setting up of photovoltaic (PV) panels which will capture solar energy and convey such energy to the convertor station in order to produce electricity exported to the national grid. Electricity in Mauritius is mainly generated from coal and heavy fuel oil, which is the baseline scenario and can also be considered as the scenario prior to the implementation of the project activity leading to considerable greenhouse gas (GHG) emissions. The project activity undertaken by project promoter Voltas Yellow Ltd will therefore substitute grid electricity by clean and renewable energy and cut down GHG emissions.

The proposed project consists of setting-up an amount of 59 376 solar PV modules with an installed capacity of 16.328 MW to produce electricity which is exported to the grid of the Central Electricity Board (CEB). The CEB in consequence will have to produce less electricity from coal or heavy fuel oil, which are non-renewable sources of energy. The energy production strategy implies a substantial reduction in the production of carbon and its release in the atmosphere, thereby reducing the associated greenhouse impact upon the atmosphere. The figure below provides an overview of the full solar PV system:



The PV modules installed are of make CSUN275-60P manufactured by China Sunergy Co.Ltd. They are of high-efficiency, poly-crystalline silicon solar cells with good temperature coefficient enabling higher output in high temperature regions, which results in module efficiency of up to 16.94%. According to the manufacturer’s warranty, the average annual power output degradation of the module shall not exceed 0.68% per year throughout the 25-years linear power output warranty, i.e. implying a life expectancy for at least this duration. For the Balance of System components, including inverters, average lifetime is estimated at 10-15 years.

In the solar photovoltaic power plant, 24 modules are connected per string of 6.6 kW output power

and connected to inverters & transformers' shelter boxes (PTR1, PTR2 & PTR4 housing 3 inverters each, while PTR3 only features 2). The solar facility thus consists of 11 Inverter Units, of make and model INGETEAM 1220TL, and 4 transformers of 440/22 kV of make and model SEA A0Bk.

A 6.5km 22KV line connects the PV plant to the substation at Riche Terre (Jin Fei). The substation (22/66KV) is the delivery and metering point, including protections and control equipment, then conveying electricity to a 14MVA ABB transformer to rise power to 22/66KV. The monitoring equipment is composed of two class 0.2 meters for import/export measurement, one of which is used for back-up, located at CEB sub-station.

The project will generate approximately 28,882 tCO₂e emission reductions per year and 202,171 tCO₂e of emission reductions over the 7 years crediting period.

During the third monitoring period 01-January-2022 to 31-December-2022, the project has operated continuously and satisfactorily generating 23.8 GWh of electrical energy and avoiding 23,557 tons of CO₂. The table below includes the audit history of the project.

Audit Type	Period	Program	VVB Name	Number of years
Validation/ Verification	19-December-2018 – 30-April-2020	VCS	Earthood Services Private Limited	1 year and 4 months
Validation/ Verification	01-May-2020 -- 31-December-2021	VCS	Carbon Check (India) Private Ltd.	1 year and 8 months
Verification	01-January-2022 -- 31-December-2022	VCS	Earthood Services Private Limited	1 year
Total				4 years

1.2 Sectoral Scope and Project Type

The sectoral scope is Scope 1 – Energy Industries (renewable sources) – ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources, Version 20.0. The project is a renewable energy type and is not a grouped project.

1.3 Project Proponent

Organization name	Voltas Yellow Ltd
Contact person	Ramchand COONJUL
Title	Solar Technical Director
Address	Beau Plan Business Park, North 1A Pamplemousses – Mauritius
Telephone	T : +230 260 5050 / M : +230 58 62 82 15
Email	rcoonjul@greenyellow.fr

1.4 Other Entities Involved in the Project

Organization name	GreenYellow SAS
Role in the Project	Corporate support
Contact person	Romane VIEIRA
Title	Global Impact Manager
Address	Tour Initiale, GreenYellow, 1, Terrasse Bellini 92800 La Défense - France
Telephone	+33 6 27 52 44 48
Email	rvieira@greenyellow.fr

1.5 Project Start Date

19-December 2018, as the date of commissioning, synchronization and start of GHG emission reductions.

1.6 Project Crediting Period

The project crediting period started on 19-December 2018 and lasts 7 years, until 18-December 2025.

1.7 Project Location

Host Party: Mauritius

Region/State/Province: District of Pamplemousses

City/Town/Community: The site is located in Solitude, which is near the village Triolet.

The project geo-coordinates are: Latitude: 20°04'44"S; Longitude: 57°32'30" E.



Figure 1: Localisation of project site

1.8 Title and Reference of Methodology

ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources, Version 20.0.

1.9 Participation under other GHG Programs

The project is registered under the Clean Development Mechanism¹¹ (Project #10483) on 31 May 2019, although its GHG emission reduction will either be claimed under the VCS programme or the CDM programme, never both.

1.10 Other Forms of Credit and Supply Chain (Scope 3) Emissions

The project does not reduce GHG emissions from activities that are included in an emissions trading program or in Scope 3 inventory emissions or any other mechanism that includes GHG allowance trading.

The project has neither sought nor received another form of GHG-related environmental credits.

1.11 Sustainable Development Contributions

The project participants are confident that the proposed project activity will make significant impact on Mauritius sustainable development. A CSR program is developed annually in order to ensure local populations' needs and to consider all our stakeholders' points of view.

In that framework, Solitude CSR program was voluntarily created and put in place since the beginning of 2022. Every year, actions with positive impacts for the Solitude community are organized and put in place. These actions are developed to respond to different needs in different areas like youth awareness, agriculture and education.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	13.0	Tons of greenhouse gas emissions avoided or removed	Implemented activities to increase	During the monitoring period, the project has allowed to avoid the emission of 23,557 tCO ₂	The project generates electricity from a renewable and clean energy source, thus avoiding the dispatch of the same amount of electricity produced by fossil-fuelled power plants to Mauritius grid. Besides, the SO _x emissions from coal-fired power plants would also be reduced.

2)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	During the third monitoring period 01-January-2022 to 31-December-2022, the project has operated continuously and satisfactorily generating 23.8 GWh of electrical energy.	The project promotes and stimulates the commercialization of grid-connected renewable energy technologies in Mauritius, in line with the Government's renewable energy policy. Adopted in 2011, the vision of Maurice Ile Durable (MID) targeted a 35% share of total energy needs to be fulfilled with renewable energies by 2025. In the Budget Speech 2021/22, the Government of Mauritius announced the Green Energy as a new economic pillar of Mauritius whereby the Renewable Energy (RE) target in the energy mix was increased from 40% to 60% by 2030 together with the announcement of the phasing out of coal in the generation of electricity by the same time. <u>(Refer to Appendix 1)</u>
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3)	7.b	7.b.1 Installed renewable energy-generating capacity in developing countries (in watts per capita)	Implemented activities to increase	No further changes this monitoring period	The project estimates to bring clean and sustainable electricity to 10,000 Mauritian households. The project will improve energy self-sufficiency of the country which is currently heavily reliant on imported fossil fuels (above 85% according to the governmental Digest of Energy and Water statistics (Ministry of Finance and Economic Empowerment, 2017), alleviating the associated risks of price variations. The government has announced plans to increase use of renewable sources of energy for electricity generation to 60 percent by 2030, and to phase out the use of coal by 2030. It aims to do this through wind farms, solar energy, biomass, wave, and waste-to-energy projects. <u>(Refer to Appendix 1)</u>
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4)	8.3	Proportion of local/regional employment opportunities	Implemented activities to increase	No further changes this monitoring period.	The project contributes to the local employment throughout its building and operation phases, its workforce consisting of up to 175 workers during the construction phase and approximately 10 to 15 workers in the operations phase (<u>Refer to Appendix 2 “Staff Log”</u>). It will also induce indirect employment by increasing the competitiveness of local industry from reducing the country’s dependency on fossil fuels. Therefore, Mauritius Government is supportive of the project because the development of solar PV power is in accordance with the national criteria for sustainable development and national policies relating to energy resources and the environment, which will push forward the use of renewable and clean energy across the country.
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6)	4.7	Provide awareness to younger generation about climate change and energy transition	Implemented activities to increase	In December 2022, 100 students were welcomed to the PV plant after having participated in a Climate Fresk Workshop to understand the stakes of global warming and how solar energy could tackle it. Climate awareness is key to the energy transition and Solitude is a means to achieve it. <u>(Refer to Appendix 3)</u>	-
7)	1.4	1.4.1 Proportion of population living in households with access to basic services	Implemented activities to decrease	A solidarity collection among GreenYellow employees was organized to donate clothes, toys and books to the association and to the children helped by it. In total, about 200 kg of donations were collected. <u>(Refer to Appendix 4)</u>	A durable partnership with Caritas Solitude – Lakaz Lespar, an organization that helps families in need, has been developed since 2020.

8)	2.4	2.4.1 Proportion of agricultural area under productive and sustainable agriculture	Implemented activities to increase	Activity carried out every year. <u>(Refer to Appendix 5)</u> No further changes this monitoring period	Since the beginning of the project, the "Petits Planteurs" cooperative comes to plant and cultivate vegetables in a more sustainable way thanks to different methods that were developed by Vijaye Chuturdharee, agricultural engineer. These plantations cover one of the four zones of the PV plant. In addition to the vegetables collection, it also helps the maintenance of the plant in an ecological way. The benefits are mutual.
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2 SAFEGUARDS

2.1 No Net Harm

An Environmental Impact Assessment (EIA) was required for the solar power plant as a “Power Generating Plant”, and it was conducted in line with the ESPA requirements. An EIA license has been obtained on September 5th, 2017 from the Ministry of the Environment and Sustainable Development for the setting-up and operation of the solar PV farm.

We have submitted eight EMRs (bi-annual reports) for the operational phase of the Photovoltaic Power Generating Plant of Capacity 16.3 MW till date, in accordance with section 18(2) (l) of the Environment Protection Act 2002 and clause 4 of the EIA licence ENV/DOE/EIA/1716 granted on 5th September 2017 for the proposed Photovoltaic Plant at Solitude.

The following key environmental and social sensitivities were considered and expected to be of negligible to minor significance in the present phase of the power generating plant:

Landscaping and top soil utilization

Areas that were cleared during construction that are not covered by the PV panels have been covered with top soil in order for vegetation to regenerate. Additionally, to avoid run-off from the water collected by the solar panels, vegetation cover is planted to capture the excess run-off. This vegetation serves to prevent erosion but also provides habitat for fauna species. Land owners are also allowed to cultivate inside the PV farm with compatible crops.

Noise

Limited transient sources of day-time transportation noise and no significant source of noise present during operations as there are no rotating machines at the PV Plant.

Air quality

There are no dust, exhaust or process emissions due to the operation of the PV Plant which affect air quality during operation period. The only source of atmospheric emissions is exhaust gas from vehicles that enter and leave the PV plant.

Water and Irrigation System

All necessary precautions are taken so that the PV development does not impact negatively on the ground water and surface water quality. Necessary measures must be taken to prevent any spillage/leakage of hazardous materials. The existing irrigation network has been left intact, with no modification and the Irrigation Authority is given free access within the perimeter of the PV farm.

Drainage system

The drainage system is also operational and the septic tank is emptied as regularly as practical. All drainage infrastructures, including soakaways have been designed and constructed to the satisfaction of the Local Authority, National Development Unit and Road Development Authority. The design of the surface drain network has been carried out such that no storm water within proposed development is channeled towards/into the drain network along the road network of the Road Development Authority.

Biodiversity

No specific biodiversity sensitivity in the Project area and well delineated Project with a limited footprint.

Waste

The project will generate significant quantities of waste during construction and decommissioning phase however these will be easily managed through industrial best practice and the waste management plan specific to the Project.

There is little domestic solid waste generated on site during the operation phase of the project. The waste generated comes from packaging materials such as plastics, cardboards and broken parts. The waste is stored in a suitable container for recycling or disposal to nearest Municipality's waste management facility. The waste transformer oil shall be sent to used-oil recyclers.

In addition to this, green waste is generated through grass cutting. The grasses are left on the ground for composting.

Landscape and visual

The natural landform of the development site is preserved and no activity related to deep excavation including rock quarrying on the plant is carried out.

Traffic

The traffic generated by the project in its current phase is limited in time and intensity as the power generating plant is not located in the busiest part of the village.

2.2 Local Stakeholder Consultation

A first Local Stakeholder Consultation (LSC) meeting was held at the site office of Voltas Yellow Ltd on 06/07/2018, with the objective of gathering comments and concerns of stakeholders on the PV Based Power Generator by Voltas Yellow Ltd in Mauritius at Solitude, District of Pamplemousses, Mauritius. The stakeholders were invited by card and by a press advert in the local newspaper. Participants were invited to give their feedback in an open and transparent manner following a presentation on the proposed project activities. The public comments received during the LSC were answered during the 1 hour meeting and no negative comment was received. A fresh LSC was not conducted during the monitoring period as there has not been any change in the project since its operation.

The LSC included:

- The representative of Voltas Yellow Ltd : Mr. Roland de Rosnay;
- The facilitator : Dr. Revin Panray Beeharry;
- Some planters/farmers;
- Some members of the District Council of Pamplemousses;
- Some members of different organisations in the locality.

The consultation meeting began with an informative presentation delivered by the facilitator on Solar PV based power generation and its role in mitigating Global Warming. He explained the phenomenon of Global Warming and the role played by Green House Gases (GHGs) and stressed on the importance of the Kyoto Protocol and the Clean Development Mechanism (CDM) as mechanism for avoiding GHG emission. Dr Revin Panray Beeharry elaborated on the climate change process and elucidated means of combating such a threat by promoting a cleaner technology through the active consideration and the adoption of 'Green Technologies' such as solar PV. Dr Revin Panray Beeharry also stressed on the direct and indirect benefits accrued by such projects and allayed fears of any environmental pollution due to the project activities. Additionally, he explained that the project would add impetus to the socio-economic fabric of the Solitude and Triolet areas and help in sustainable development, as promulgated by the Kyoto Protocol and United Nation Framework Convention on Climate Change.

Stakeholder Consultation Meeting

Attendance Sheet

Date: Friday 06 July 2018

Time: 15:00

Venue: On the site of the project at Solitude opposite the Football Ground

No	Name	Name of Organisation	Designation	Signature	Tel/Mob
1	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
2	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
3	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
4	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
5	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
6	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
7	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
8	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
9	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
10	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
11	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
12	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
13	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
14	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109

No	Name	Name of Organisation	Designation	Signature	Tel/Phone
15	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
16	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
17	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
18	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
19	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
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28	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
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31	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
32	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109
33	Dr. Revin Panray Beeharry	Dr. Revin Panray Beeharry	Dr. Revin	[Signature]	52602109



During the Question and Answer session, it was more clear to the audience as to how the PV project was less polluting and had no side effects compared to conventional ways of power generation. People also exchanged their views on the changes and benefits brought to the village, by the project. The audience looked forward to such good development projects and pondered upon how such projects would fare in the future and whether this project would stand the test of time and last for at least 20 years.

Stakeholders in general expressed their satisfaction at the initiative and welcomed the project. From their perspective, the project activity would accelerate socio-economic development in the area.

During this monitoring period 01–January 2022 to 31–December 2022, no formal consultation meeting was held and no other local stakeholder comments were received but regular contact has been maintained with Mr. Vijaye Chuturdharee over the year.

Throughout the operation phase of the solar farm, two official environmental reports are prepared on a bi-annual basis which are also transferred to the stakeholders. This tool allows the stakeholders to follow the project's development and raise their queries, if they have any. A log sheet is also available on site whereby visitors and stakeholders are encouraged to register their remarks and grievances, if they have any.

Regular exchanges take place with farmers belonging to the cooperative “Les Petits Planteurs”. Every year, they come and cut the grass that grows under the solar panels and take it away to feed their livestock.

2.3 AFOLU-Specific Safeguards

Not applicable (this project is not an AFOLU project)

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The commission date of the project activity is 19-December 2018, which is also the start date of GHG emission reductions. During the monitoring period 01–January 2022 to 31–December 2022, the project has operated continuously and satisfactorily, generating a total net energy delivered to the grid equal to 23,8 GWh. The only noteworthy event during the monitoring period involved the power phase damages of the inverters which led to under-performance. The root cause of the power phase damage was identified as a manufacturing defect. No changes on the project activity or applicable legislation have occurred during this monitoring period.

3.2 Deviations

3.2.1 Methodology Deviations

No methodology deviations occurred during this monitoring period.

3.2.2 Project Description Deviations

No methodology deviations occurred during this monitoring period.

3.3 Grouped Projects

This is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EFgrid,CM,y
Data unit	tCO ₂ e/MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year
Source of data	Representatives of Ministry of Energy of Mauritius
Value applied	0.9915
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EFgrid,OM,y
Data unit	tCO ₂ e/MWh
Description	Operating Margin CO ₂ emission factor for grid connected power generation in year y calculated using the CDM "Methodological Tool to calculate the emission factor for an electricity system".
Source of data	IGES List of Grid Emission Factors
Value applied	1.0282
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EFgrid,BM,y
Data unit	tCO ₂ e/MWh
Description	Build Margin CO ₂ emission factor for grid connected power generation in year y calculated using the CDM "Methodological Tool to calculate the emission factor for an electricity system".

Source of data	IGES List of Grid Emission Factors
Value applied	0.8814
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	The percentage share of total installed capacity of solar PV
Data unit	%
Description	The percentage share of total installed capacity of the solar PV in the total installed grid connected power generation capacity in the host country.
Source of data	http://publicutilities.govmu.org/
Value applied	2.6
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Additionality demonstration
Comments	-

Data / Parameter	The total installed capacity of solar PV
Data unit	MW
Description	The total installed capacity of grid-connected solar PV in the host country.
Source of data	http://publicutilities.govmu.org/
Value applied	21.2
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Additionality demonstration
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	EGfacility,y																							
Data unit	MWh/yr																							
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y																							
Source of data	Measured directly with electricity meter(s) at Interconnection Facilities CEB JINFEI Substation																							
Description of measurement methods and procedures to be applied	-																							
Frequency of monitoring/recording	Continuous measurement and at least monthly recording																							
Value monitored	<table><tr><td>2022</td><td>Total</td></tr><tr><td>23,764</td><td>23,764</td></tr></table>			2022	Total	23,764	23,764																	
2022	Total																							
23,764	23,764																							
Monitoring equipment	Electricity outputs are electronically recorded, stored and invoiced monthly to CEB by Voltas Yellow Ltd, based on main meter below (or back-up in case of main meter unavailability or default). The measurement devices of net electrical energy delivered to CEB grid are both located 20/63kV at JINFEI substation and feature the following specifications: <table><tr><td>Meter designation:</td><td>Main</td><td>Back-up</td></tr><tr><td>Meter ownership:</td><td>CEB</td><td>CEB</td></tr><tr><td>Make:</td><td>EDMI</td><td>EDMI</td></tr><tr><td>Model:</td><td>MK6E</td><td>MK6E</td></tr><tr><td>Accuracy class:</td><td>0.2S (±0.02%)</td><td>0.2S (±0.02%)</td></tr><tr><td>Serial Number:</td><td>16538506</td><td>16538507</td></tr><tr><td>Calibration date:</td><td>22.06.2020</td><td>22.06.2020</td></tr></table>			Meter designation:	Main	Back-up	Meter ownership:	CEB	CEB	Make:	EDMI	EDMI	Model:	MK6E	MK6E	Accuracy class:	0.2S (±0.02%)	0.2S (±0.02%)	Serial Number:	16538506	16538507	Calibration date:	22.06.2020	22.06.2020
Meter designation:	Main	Back-up																						
Meter ownership:	CEB	CEB																						
Make:	EDMI	EDMI																						
Model:	MK6E	MK6E																						
Accuracy class:	0.2S (±0.02%)	0.2S (±0.02%)																						
Serial Number:	16538506	16538507																						
Calibration date:	22.06.2020	22.06.2020																						
QA/QC procedures to be applied	Cross check of measurement results with records for sold electricity. Meter Laboratory of CEB is solely responsible for the selection, installation, calibration, servicing, testing and repairing of all CEB energy meters (CEB Main Meter & CEB Back-up Meter). Voltas Yellow Ltd may install at its own expense a backup electric metering device (Seller Back-up Meter). As per PPA §12.1.4, CEB shall inspect each CEB Meter upon installation and at least once every year thereafter. CEB shall																							

	check the certification of CEB Meters through an accuracy test at least once every 4 (four) years thereafter or at any time the readings of Net Energy from the CEB Meter and Seller Back-up Meter differ by an amount greater than 0.5%.
Purpose of the data	Calculation of baseline emissions
Calculation method	-
Comments	The energy meters at the Jin Fei substation meters have not been inspected and load tests have not been carried out on the meters to confirm their correctness, as per schedule 12.1.4 of the Energy Supply and Purchase Agreement during the monitoring period. It was agreed by CEB and Voltas Yellow Ltd that the accuracy tests would take place in the second half of 2022, during an operational shut down of the farm. Nonetheless, this didn't happen in 2022 and as a result, the accuracy tests were planned at the earliest, the following year. Hence the latest accuracy tests have been carried out in February 2023.

4.3 Monitoring Plan

The proposed project activity monitoring plan complies with the methodology ACM0002 Grid-connected electricity generation from renewable sources, Version 20.0 whereby it is stated that: As per ACM002 provisions for record handling, all data collected as part of monitoring is archived electronically and kept at least for 2 years after the end of the last crediting period. All measurements are conducted with calibrated measurement equipment according to relevant industry.

Indeed, the quantity of net electricity generation supplied by the project plant to the grid is reliably monitored through calibrated electricity meters and cross-checked with sales records as part of quality assurance/quality control measures on top of CEB diligence. Besides, the reading of the main meter is compared with the check meter at the point of injection.

Data is also daily monitored, analyzed and cross-checked by the monitoring team through Quantum (QOS Energy) which is a web-based energy management platform that is in real-time communication with the equipment on site. The data acquisition equipment and installation are subject to regular maintenance and verifications as per the equipment manuals and the meters, to testing as per the ESPA, as stipulated in section 4.

Monitoring organization:

The Operations Director of Green Yellow Indian Ocean coordinates and endorses the overall responsibility for all CDM monitoring of the project, including:

- Develop, approve, execute, and improve the CDM Monitoring/Reporting Procedures;
- Organize in-house seminar to inform and train the company staff to the monitoring procedures;
- Ensure that instrumentations and devices are available and properly suited to efficiently perform the monitoring;
- Communicate and coordinate the monitoring work of all business units;

- Validate and electronically archive all monitoring data on a monthly basis throughout the crediting period (and conserve it at least for 2 further years);
- Calculate and report the emission reductions; and
- Coordinate the DOE work during the verification audit.

The Director might appoint a CDM coordinator to delegate to him the above specific tasks of monitoring supervision. The Technical/Engineering/Maintenance Department consisting of plant technicians will undertake the technical actions required by the monitoring plan, under the Directors' authority, to collect and record related data.

The Accounting/Sales Department (Chief Financial Officer) will crosscheck, reconcile or consolidate data with multiple sources whenever possible. At minimum, data obtained from the electricity meters is to be crosschecked with the electricity sales receipts. This kind of reconciliation activity will be recorded properly as DOE may request for such information during the verification.

Data collection, consolidation and results analysis is undertaken by a dedicated team adequately trained, well aware of CDM requirements. O&M workflow manual enforced on-site further includes all updated procedures in case of mistake/break-downs, emergency disconnections and cyclone warnings.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The monitoring period for which GHG emission reductions were achieved spans from 01/01/2022 to 31/12/2022. Baseline emissions are calculated according to the methodology as the product of (i) the quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y ($EG_{PJ,y}$ = Exports – Imports, in MWh/yr) and (ii) the combined margin CO₂ emission factor for grid connected power generation in year y ($EF_{grid,CM,y}$):

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y	Baseline emissions in year y (t CO ₂ /yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO ₂ /MWh)

$EG_{PJ,y} = EG_{facility}$ i.e. the quantity of net electricity generation supplied by the project plant to the grid, as monitored and displayed below on a monthly basis.

$EF_{grid,CM,y} = 0.9915 \text{ tCO}_2/\text{MWh}$, as the Project Proponent have calculated, the DOE validated and the CDM-EB registered.

Year	Imports	Exports	$EG_{facility,y}$	$EF_{grid,CM,y}$	BE_y
	kWh	kWh	MWh	tCO ₂ /MWh	tCO ₂
Jan-2022	9,009	2,198,673	2,189	0.9915	2,170
Feb-2022	7,551	1,847,589	1,840	0.9915	1,824
Mar-2022	9,213	1,931,508	1,922	0.9915	1,905
Apr-2022	9,620	1,870,941	1,861	0.9915	1,845
May-2022	10,212	1,892,187	1,882	0.9915	1,866
June-2022	10,100	1,592,934	1,582	0.9915	1,568
July-2022	10,355	1,827,403	1,817	0.9915	1,801
Aug-2022	10,237	1,928,686	1,918	0.9915	1,901
Sept-2022	9,663	2,123,942	2,114	0.9915	2,096
Oct-2022	9,606	2,193,839	2,184	0.9915	2,165
Nov-2022	8,725	2,221,665	2,213	0.9915	2,194
Dec-2022	8,560	2,250,811	2,242	0.9915	2,222
TOTAL	112,851	23,880,178	23,764		23,557

5.2 Project Emissions

No project emissions are expected as the project activity only involves renewable electricity generation from the solar power plant: $PE_y = 0$.

5.3 Leakage

As stated in the applicable methodology, no leakage emissions are considered.

5.4 Net GHG Emission Reductions and Removals

The monitoring period for which GHG emission reductions were achieved spans from 01/01/2022 to 31/12/2022, as it follows:

Year	Baseline emissions or removals(tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals(tCO ₂ e)
2022	23,557	0	0	23,557
Total	23,557			23,557

<u>Ex-ante emissions reductions/removals</u>	<u>Achieved emissions reductions/removals</u>	<u>Percent difference</u>	<u>Justification for the difference</u>
28,978 tCO ₂ e	23,557 tCO ₂	19%	The difference can be explained by : i. overall solar PV lower performance due to the power phase damage which was identified as a manufacturing defect. ii. Temperature losses, weaker annual global solar irradiation, shadings, dust and cable transmission losses.

6 APPENDICES

Appendix 1: Electricity generation invoices for January, April, August and December 2022



Voltas Yellow Ltd
10, Louis Pasteur Street,
Port Louis
Mauritius
BRN: C16137562
VAT: 27393506

Invoice date: 04/02/2022

INVOICE TO:
The Central Electricity Board
Royal Road
Curpipe
Mauritius
VAT: 22000591
BRN F07000041

VAT INVOICE

INVOICE NUMBER VY 2022/01/02.04

DESCRIPTION	AMOUNT (MUR)
BILLING FOR SOLITUDE SOLAR FARM 16.3 MWc - January 2022	
Calculations based on energy data provided by CEB on 04/02/2022	
Energy index on 01/02/2022	kWh 76 307 525
Energy index on 01/01/2022	kWh 74 108 852
NE - Net Energy*	kWh 2 198 673
DE = Deemed Energy	kWh 0,00
SE = Standard Energy (NE+DE)	kWh 2 198 673
EE = Excess Energy	kWh -
T _e	MUR/kWh 0,795
T _g	USD/kWh 0,088
FX _{MUR/USD}	as at 31/01/2022 43,150
Standard Tariff (ST ₁)	MUR/kWh 4,59200
Excess Tariff (ET ₁)	MUR/kWh 0,45920
Monthly Energy Charge	=(SE*ST ₁) + (EE*ET ₁)
	MUR (excl. VAT) 10 096 746,15
	MUR VAT (15%) 1 514 511,92
	MUR (incl. VAT) 11 611 258,07

Terms of payment: Bank transfer 100% in MUR
Deadline of payment: 06/03/2022
Account Name: VOLTAS YELLOW LTD
Bank: BCP Bank (Mauritius)
Account No.: 20126111000
Currency: MUR

Francois-Hugues D'Allières
CFO



Voltas Yellow Ltd
10, Louis Pasteur Street,
Port Louis
Mauritius
BRN: C16137562
VAT: 27393506

VAT INVOICE

INVOICE NUMBER VY 2022/04/05.04

DESCRIPTION	AMOUNT (MUR)
BILLING FOR SOLITUDE SOLAR FARM 16.3 MWc - April 2022	
Calculations based on energy data provided by CEB on 04/05/2022	
Energy index on 01/05/2022	kWh 81 957 563
Energy index on 01/04/2022	kWh 80 066 421
NE - Net Energy*	kWh 1 870 941
DE = Deemed Energy	kWh 0,00
SE = Standard Energy (NE+DE)	kWh 1 870 941
EE = Excess Energy	kWh -
T _e	MUR/kWh 0,795
T _g	USD/kWh 0,088
FX _{MUR/USD}	as at 29/04/2022 42,975
Standard Tariff (ST ₁)	MUR/kWh 4,576800
Excess Tariff (ET ₁)	MUR/kWh 0,457680
Monthly Energy Charge	=(SE*ST ₁) + (EE*ET ₁)
	MUR (excl. VAT) 8 562 922,77
	MUR VAT (15%) 1 284 838,42
	MUR (incl. VAT) 9 847 761,18

Terms of payment: Bank transfer 100% in MUR
Deadline of payment: 05/06/2022
Account Name: VOLTAS YELLOW LTD
Bank: BCP Bank (Mauritius)
Account No.: 20126111000
Currency: MUR

Francois-Hugues D'Allières
CFO



Voltas Yellow Ltd
10, Louis Pasteur Street,
Port Louis
Mauritius
BRN: C16137562
VAT: 27393506

Invoice date: 06/09/2022

INVOICE TO:
The Central Electricity Board
Royal Road
Curpipe
Mauritius
VAT: 22000591
BRN F07000041

VAT INVOICE

INVOICE NUMBER VY 2022/08/09.06

DESCRIPTION	AMOUNT (MUR)
BILLING FOR SOLITUDE SOLAR FARM 16.3 MWc - August 2022	
Calculations based on energy data provided by CEB on 05/09/2022	
Energy index on 01/09/2022	kWh 89 198 773
Energy index on 01/08/2022	kWh 87 270 087
NE - Net Energy*	kWh 1 928 686
DE = Deemed Energy	kWh 0,00
SE = Standard Energy (NE+DE)	kWh 1 928 686
EE = Excess Energy	kWh -
T _e	MUR/kWh 0,795
T _g	USD/kWh 0,088
FX _{MUR/USD}	as at 31/08/2022 44,350
Standard Tariff (ST ₁)	MUR/kWh 4,697800
Excess Tariff (ET ₁)	MUR/kWh 0,469780
Monthly Energy Charge	=(SE*ST ₁) + (EE*ET ₁)
	MUR (excl. VAT) 9 060 581,09
	MUR VAT (15%) 1 359 087,16
	MUR (incl. VAT) 10 419 668,25

Terms of payment: Bank transfer 100% in MUR
Deadline of payment: 06/10/2022
Account Name: VOLTAS YELLOW LTD
Bank: BCP Bank (Mauritius)
Account No.: 20126111000
Currency: MUR

Francois-Hugues D'Allières
CFO



Voltas Yellow Ltd
10, Louis Pasteur Street,
Port Louis
Mauritius
BRN: C16137562
VAT: 27393506

VAT INVOICE

INVOICE NUMBER VY 2022/12/01.10

DESCRIPTION	AMOUNT (MUR)
BILLING FOR SOLITUDE SOLAR FARM 16.3 MWc - December 2022	
Calculations based on energy data provided by CEB on 10/01/2023	
Energy index on 01/01/2023	kWh 97 989 030
Energy index on 01/12/2022	kWh 95 738 219
NE - Net Energy*	kWh 2 250 811
DE = Deemed Energy	kWh 0,00
SE = Standard Energy (NE+DE)	kWh 2 250 811
EE = Excess Energy	kWh -
T _e	MUR/kWh 0,795
T _g	USD/kWh 0,088
FX _{MUR/USD}	as at 30/12/2022 43,660
Standard Tariff (ST ₁)	MUR/kWh 4,636200
Excess Tariff (ET ₁)	MUR/kWh 0,463620
Monthly Energy Charge	=(SE*ST ₁) + (EE*ET ₁)
	MUR (excl. VAT) 10 435 209,96
	MUR VAT (15%) 1 565 281,49
	MUR (incl. VAT) 12 000 491,45

Terms of payment: Bank transfer 100% in MUR
Deadline of payment: 07/02/2023
Account Name: VOLTAS YELLOW LTD
Bank: BCP Bank (Mauritius)
Account No.: 20126111000
Currency: MUR

Francois-Hugues D'Allières
CFO

Appendix 2: "Employment Log"

Solitude PV Farm - Operations Phase - Employees/subcontractors log
Contact number : 260 5050
Date : 13/07/2023



No.	Name	Employee/Subcontractor involved	Activity
1	Ramchand Coorjaj	GreenYellow Indian Ocean	O&M
2	Justin Arlanda	GreenYellow Indian Ocean	O&M
3	Ricardo Sandian	GreenYellow Indian Ocean	O&M
4	Rivona Ramasampurnoj	GreenYellow Indian Ocean	Supervision
5	Herinjala Andrianarivo	GreenYellow Indian Ocean	Supervision
6	Lovisa Rason	GreenYellow Indian Ocean	Supervision
7	Harizo Rakotonmalala	GreenYellow Indian Ocean	Supervision
8	Paul Andrihoaby	GreenYellow Indian Ocean	Supervision
9	Virgil Sredoo	GreenYellow Indian Ocean	O&M
10	Opika Callycham	GreenYellow Indian Ocean	O&M
11	Marion Michot	GreenYellow Indian Ocean	HSE & CSR
12	Lumitech		Annual Maintenance
13	DYN AGRO Energy Consulting & Service Provider Ltd		Weed Control & Grass Maintenance
14	CAM ACTIVE		Security Service


Technical Director - PV O&M & IRVE
GreenYellow Indian Ocean




Appendix 3:



Appendix 4:



Appendix 5:



Mati Ke Mol - Agri Solar Farming - YouTube : Link to documentary